Install DHCP Server on Windows Server

1️⃣ Install DHCP Feature

Open `PowerShell` as Administrator and run:

```
Install-WindowsFeature DHCP -IncludeManagementTools
```

Create and Configure DHCP Scope

1️⃣ Create new IPv4 scope

```
Add-DhcpServerv4Scope -Name "3bhits" -StartRange 10.0.5.50 -EndRange 10.0.5.150 -SubnetMask 255.255.255.0 -State Active
```

2️⃣ List all scopes

```
Get-DhcpServerv4Scope
```

> **Important**
>
> If nothing is in the list, there was an error creating the scope!

3️⃣ Configure DNS server (Option 6)

```
Set-DhcpServerv4OptionValue -ScopeId 10.0.5.0 -OptionId 6 -Value "8.8.8.8"
```

4️⃣ Configure Default Gateway (Option 3)

```
Set-DhcpServerv4OptionValue -ScopeId 10.0.5.0 -OptionId 3 -Value "10.0.5.2"
```

5️⃣ List all configured option values

```
Get-DhcpServerv4OptionValue -ScopeId 10.0.5.0
```

> **Note**
>
> You should see at least 2 configured options!

⚠️ Remove Scope (if you need to)

```
Remove-DhcpServerv4Scope -ScopeId 10.0.10.0
```

> **Important**
>
> Replace the `-ScopeId` with the one you want to delete!

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Create DHCP Reservation

1️⃣ Create reservation using existing lease

This command automatically uses the `ClientId` from the current lease:

```
Add-DhcpServerv4Reservation -ScopeId 10.0.5.0 -ClientId (Get-DhcpServerv4Lease -ScopeId 10.0.5.0).ClientId -IPAddress 10.0.5.25 -Name "debianClient1"
```

> **Note**
> This only works if the client has already received an IP address from the DHCP server

2️⃣ Add reservation manually for a specific client (❗ optional ❗)

```
Add-DhcpServerv4Reservation -ScopeId 10.0.5.0 -IPAddress 10.0.5.51 -ClientId "00-11-22-33-44-55"
```

> **Important**
>
> The `-IPAddress` must be within your configured DHCP scope
> The `-ClientId` must be the MAC address of the client device

Test Connection on Linux Client

1️⃣ Test connectivity

`ping google.com`

🎉 Congratulations! You're all set

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